Narin Dhatwalia

Sr. Technical Architect, AI/ML
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Summary

Technical Architect with 6+ years building and deploying enterprise AI/ML systems for Fortune 500 clients, combining production machine learning, SQL-based data work, and stakeholder-focused analytics. At Snowflake, leads customers from discovery to production on LLM applications, agents, and predictive models, including a clinical-data pipeline analyzing ~400,000 patients that projected ~$34M in new revenue and supported a $7.5M contract renewal. Won first prize for a fine-tuning approach now being productized, with an MS in Machine Learning and Data Science from Northwestern and ongoing DBA research on enterprise GenAI's financial impact.

Work Experience

Snowflake | ***Sr. Technical Architect, AI/ML*** **Jun 2024 - Present**

- Won first prize at an internal Snowflake hackathon by fully fine-tuning an open-weight model (Qwen3-8B) on GPUs using responses from a frontier-lab teacher LLM. Compared the two on an unseen test set for latency, cost, and quality: lower cost and latency with comparable quality. Leadership and Product & Engineering have backed it as a native fine-tuning experience for customers, and I was selected to present to the CEO.
- Architected a production Cortex AI pipeline leveraging Claude 3.5 Sonnet to analyze clinical data for ~400,000 patients and recommend HCC billing codes with supporting clinical reasoning, projecting ~$34M in new revenue and supporting a $7.5M contract renewal. Built an LLM evaluation framework for accuracy, groundedness, and context relevance to monitor outputs.
- Designed a Snowflake CoWork data product for a financial services client, replacing manual SQL workflows with a unified Cortex Agent that combined text-to-SQL over a bespoke semantic layer with search over unstructured CRM notes, returning answers in seconds. Instrumented the solution with experimental-design-based evals and AI observability covering cost, latency, user feedback, prompt and response traces; enforced governance with Row Access Policies; and worked with Snowflake Product & Engineering on private preview features. Defined the DEV-to-PROD promotion gate for agents with an LLM-as-judge eval suite and pass/fail criteria, replacing subjective spot checks.
- Built ML models across industries: an XGBoost router outage prediction model with time-series cross-validation for a telecommunications client, reducing network downtime by 40%; a new-release revenue prediction model for a media client; and a transaction categorization model for a fintech client covering ~50M payment records weekly. The telecommunications proof of concept contributed to a $7.2M Snowflake deal.
- Built a text-to-SQL chatbot with Cortex Analyst for a major SaaS company, enabling Customer Success Managers to query account health data in natural language and reducing SQL bottlenecks by 80%.

TransUnion | ***Consultant, Data Science & Analytics*** **Jan 2023 - Jun 2024**

- Built attribute sets (predictive features of consumer creditworthiness) feeding originations-risk, account management, and marketing propensity models; owned "Soft Inquiry Attributes" end to end (SQL/Spark extraction, R feature calculation) and measured performance lift in credit card marketing models.
- Led a 6-month Data Science Innovation study comparing GPU-accelerated Random Forest against the XGBoost-based originations baseline: significantly lower training time with less overfitting.
- Developed an originations-risk underwriting model using Survival Analysis that predicts the month of potential delinquency, raising AUC from 0.80 to 0.84 and outperforming legacy models.

Decision Point Analytics | ***Data Scientist*** **Feb 2020 - Jun 2021**

- Built Bayesian regression Market Mix Models for Kellogg's LATAM to allocate marketing spend across Digital, TV, and Print, incorporating expert input through conjugate priors.
- Used Facebook's Prophet to optimize Coca-Cola Mexico's promotions, decomposing volume lift into forward buying, cannibalization, and true lift, driving a 20% revenue boost the following quarter.

Antuit.ai | ***Business Analyst*** **Dec 2017 - Sep 2018**

- Built automated ETL pipelines across Nielsen, NPD, and IRi data plus Tableau/Power BI dashboards, cutting turnaround by 6 days.

Projects

Text-to-SQL CLI Agent | GitHub portfolio

- Built a conversational data product for querying any SQLite database with SQL, using an open-weight model served by Fireworks AI, schema grounding, execution-guided self-repair, and a two-layer guardrail. Evaluated by executing generated queries and comparing result rows, achieving 9/10 on its evaluation set at 1.15s median latency versus 1/10 for a naive prompt on the same model.

Generative AI: Infinite Shakespeare with Transformers | GitHub portfolio

- Built and trained a decoder-only Transformer language model in PyTorch on the complete works of Shakespeare, implementing multi-head self-attention, feed-forward blocks, skip connections, layer normalization, and triangular masking for token-level affinities to generate an endless stream of text resembling his plays.

Education

Ottawa University | ***Doctorate, Business Administration*** **Dec 2025 - Present**

Northwestern University | ***MS, Machine Learning and Data Science*** **(GPA: 3.926)** **Sep 2021 - Dec 2022**

University of Delhi | ***BS (Honors), Mathematics*** **(GPA: 81.46%)** **Jul 2014 - Jul 2017**

Skills

- **Languages and Data**: SQL (Snowflake, PostgreSQL), Python, R, Spark (distributed computing), pipeline orchestration
- **ML and AI**: Scikit-learn, XGBoost, PyTorch, TensorFlow, LangChain, LangGraph, RAG, agentic architectures (tool integrations, guardrails), MCP integrations, LLM evaluation (LLM-as-judge), fine-tuning and distillation, open-weight serving (Fireworks AI), Experimental design
- **Snowflake (SnowPro Core Certified)**: Cortex AI (Agents, Search, Analyst), Snowflake CoWork, Snowpark, UDFs/UDTFs, Streamlit, Notebooks, Model Registry, Feature Store, Dynamic Tables, RBAC, Stored Procedures
- **BI and Databases**: Tableau, Power BI, Sigma, Neo4j (Graph Database), BigQuery, Hex
- **AWS**: S3, RDS, ECS, ECR, SageMaker, EC2, VPC, CloudFormation, Route53, AWS CLI
- **Data Science and Business**: B2B Enterprise Sales Metrics, Payments and Financial Technology Domain, Data Product Development
- **Development Practices**: Version Control